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Studierichting: Informatica
Oefeningenreeks 5G nr. 6.6

$$f(x) = -2x^2 + 6x$$

$$g(x) = x^2$$

$$f(x) = g(x)$$

$$-2x^2 + 6x = x^2$$

$$\Leftrightarrow -3x^2 + 6x = 0$$

$$\Leftrightarrow -3x(x - 2) = 0$$

$$\Leftrightarrow x = 0 \text{ of } x = 2$$

$$\Rightarrow x \in [0, 2]$$

We testen een punt tussen 0 en 2: $x = 1$

$$f(1) = -2(1)^2 + 6(1) = 4$$

$$g(1) = 1^2 = 1$$

$\Rightarrow f(x)$ is de bovenste functie

$$\Rightarrow V = \pi \int_0^2 (f(x)^2 - g(x)^2) dx$$

$$= \pi \int_0^2 ((-2x^2 + 6x)^2 - (x^2)^2) dx$$

$$(-2x^2 + 6x)^2 = 4x^4 - 24x^3 + 36x^2$$

$$(x^2)^2 = x^4$$

$$\Rightarrow (-2x^2 + 6x)^2 - (x^2)^2 = 4x^4 - 24x^3 + 36x^2 - x^4$$

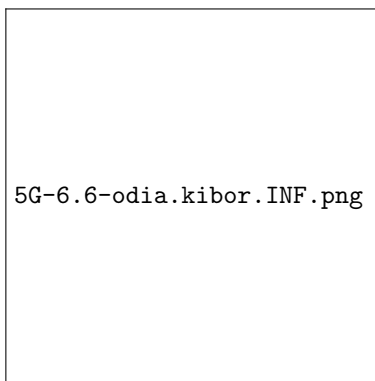
$$= 3x^4 - 24x^3 + 36x^2$$

$$\Rightarrow V = \pi \int_0^2 (3x^4 - 24x^3 + 36x^2) dx$$

$$= \pi \left[\frac{3x^5}{5} - \frac{24x^4}{4} + \frac{36x^3}{3} \right]_0^2$$

$$= \pi \left[\frac{3x^5}{5} - 6x^4 + 12x^3 \right]_0^2$$

$$\begin{aligned}
&= \pi \left(\frac{3(2)^5}{5} - 6(2)^4 + 12(2)^3 \right) - \pi \left(\frac{3(0)^5}{5} - 6(0)^4 + 12(0)^3 \right) \\
&= \pi \left(\frac{96}{5} - 96 + 96 \right) - \pi(0) \\
&= \pi \cdot \frac{96}{5} \\
\Rightarrow V &= \frac{96\pi}{5}
\end{aligned}$$



References